FG DEGREE COLLEGE ABID MAJEED ROAD RAWALPINDI AFFILIATED WITH PUNJAB UNIVERSITY

PSYCHOPATHOLOGY

SUBMITTED BY:

- **SADIA SHAFIQUE**

SUBMITTED To:

- **MAM KHADIJA**

DEPARTMENT:

- **BS PSYCHOLOGY 5TH SEMESTER**

DATE:

- **21st April,2024**

TOPIC:

- **NEUROCOGNITIVE DISORDERS**

- **PSYCHOPATHOLOGY:**
  Psychopathology is the scientific exploration of abnormal mental states that, for more than a century, has provided a Gestalt for psychiatric disorders and guided clinical as well as scientific progress in modern psychiatry.
- **NEUROCOGNITIVE DISORDERS:**
  Neurocognitive disorder is a general term that describes decreased mental function due to a medical disease other than a psychiatric illness.

  Neurocognitive disorders are grouped into three subcategories:

  1. **Delirium.**
  2. **Mild neurocognitive disorder – some decreased mental function, but able to stay independent and do daily tasks.**
  3. **Major neurocognitive disorder – decreased mental function and loss of ability to do daily tasks. Also called dementia.**
- **DELIRIUM:**
  A mental state in which a person is confused and has reduced awareness of their surroundings. The person may also be anxious, agitated, or have less energy than usual and be tired or depressed.

  DSM-5 Diagnostic Criteria :

  A. **A disturbance in attention (i.e., reduced ability to direct, focus, sustain, and shift Attention) and awareness (reduced orientation to the environment).**

B. The disturbance develops over a short period of time (usually hours to a few days), Represents a change from baseline attention and awareness, and tends to fluctuate During the course of the day.
C. An additional disturbance in cognition (e.g., memory deficit, disorientation, language, visuospatial ability, or perception).

D. These cognitive disturbances are not better explained by another pre-existing, established, or evolving neurocognitive disorder and do not occur in the context of a Severely reduced level of arousal, such as coma.
E. There is evidence from the history, physical examination, or laboratory findings that The disturbance is a direct physiological consequence of another medical condition, Substance intoxication or withdrawal (i.e., due to a drug of abuse or to a medication), Or exposure to a toxin, or is due to multiple etiologies.

- **Other specified delirium:** as for delirium but symptoms do not meet the full criteria but Cause clinically significant distress or impairment.
  NOTE: The clinician must specify the reason the delirium does not meet the full criteria.
- **Unspecified delirium:** as for delirium but symptoms do not meet the full criteria but cause Clinically significant distress or impairment.
  NOTE: The clinician does not specify the reason the delirium does not meet the full Criteria.

- **Major Neurocognitive Disorder:**
  Major neurocognitive disorder is characterized by a significant decline in at least one of the domains of cognition which include executive function, complex attention, language, learning, memory, perceptual-motor, or social cognition.

  DSM-5 Diagnostic Criteria :

  A. Evidence of significant cognitive decline from a previous level of performance in one or more cognitive domains (complex attention, executive function, learning and memory, language, perceptual-motor, or social cognition) based on:
  1. Concern of the individual, a knowledgeable informant, or the clinician that there has been a significant decline in cognitive function; and
  2. A substantial impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment.

B. The cognitive deficits interfere with independence in everyday activities (i.e., at a minimum, requiring assistance with complex instrumental activities of daily living such as paying bills or managing medications).

C. The cognitive deficits do not occur exclusively in the context of a delirium.
D. The cognitive deficits are not better explained by another mental disorder (e.g., major depressive disorder, schizophrenia).

- **MILD NEUROCOGNITIVE DISORDER:**
  Mild neurocognitive disorder is a condition characterized by a slight decline in cognitive abilities, such as memory, language, or problem-solving skills, which is noticeable but does not significantly interfere with daily functioning or independence.

  DSM-5 DIAGNOSTIC CRITERIA:
  DSM-5 mild neurocognitive disorder has the same criteria as for major neurocognitive disorder, except that the cognitive decline is modest/mild and does not interfere with capacity for independence in everyday activities.
  The reliability and validity of this diagnosis in individuals with ID has not been established and we advise caution in using this diagnosis.

- **MAJOR AND MILD NEUROCOGNITIVE DISORDERS DUE TO ALZHEIMER'S DISEASES:**

  - **ALZHEIMER'S DISEASE:**
    Alzheimer's disease is the most common type of dementia. It is a progressive disease beginning with mild memory loss and possibly leading to loss of the ability to carry on a conversation and respond to the environment. Alzheimer's disease involves parts of the brain that control thought, memory, and language.

AYESHA SHERAZ

"2186"

Major or Mild Neurocognitive disorder due to Alzheimer's disease

Alzheimer's disease is a brain disorder that slowly destroys memory and thinking skills, and eventually, the ability to carry out the simplest tasks

- **Diagnostic criteria:**

A. Criteria are met for major and mild neurocognitive disorder.

B. Onset and gradual progression of impairment, at least two cognitive domains.

C. Criteria met for either probable or possible Alzheimer's disease as fellows:

- **For Major Neurocognitive disorder:**

Probable Alzheimer's disease is diagnostic if either of the following is present; otherwise, Possible Alzheimer's disease should be diagnosed.

1) Causative Alzheimer's genetic mutation from family history or genetic testing.

2) All Three of the following are present:

a) Decline in cognition, memory and learning

b) Steadily progressive, without extended plateaus

c) No evidence of mixed etiology

- **For Mild neurocognitive disorder:**

Probable Alzheimer's disease is diagnosed if there is evidence of a causative Alzheimer's genetic mutation from family history or genetic testing.

Possible Alzheimer's disease is diagnosed if there is no evidence of a causative Alzheimer's genetic mutation from family history or genetic testing and All Three of the following are present:

a) Decline in cognition, memory and learning

b) Steadily progressive, without extended plateaus

c) No evidence of mixed etiology

D.Thedisturbanceisnotbetterexplainedbyanotherneurocognitivediseaseoranothermental disorder.

Major or Mild Frontotemporal Neurocognitive disorder

It is the mental health condition characterized by abnormal shrinkage in two parts of the brain, called the frontal temporal anterior lobes, later in DSM-5 called it Frontotemporal dementia.

- **Diagnostic criteria:**

A. Criteria are met for major and mild neurocognitive disorder.

B. Disturbance has insidious onset and gradual progression

C. Either (1) 0r (2):

1. Behavioral variant 2. Language variant

1. Behavioral variant:

A) Three or more of the following behavioral symptoms:

I. Behavioral disinhibition

II. Apathy or inertia

III. Loss of sympathy or empathy

IV. Perseverative, stereotyped or compulsive behavior

V. Hyper orality and dietary changes

B) Prominent decline in social cognition or executive abilities.

2. Language variant:

A) Prominent decline in language ability, in the form of speech production, word finding, object naming, grammar, or word comprehension.

D. Sparing of learning and memory and perceptual-motor function.

E. The disturbance is not better explained by another neurocognitive disease or another mental disorder.

Probable frontotemporal neurocognitive disorder is diagnostic if either of the following is present; otherwise, possible frontotemporal neurocognitive disorder should be diagnosed:

1) Causative frontotemporal genetic mutation, from either family history or genetic testing.

2) Evidence of disproportionate frontal or temporal lobe involvement from neuroimaging.

Major or Mild Neurocognitive disorder with Lewy Bodies

Lewy Bodies = Microscopic proteins deposits that damage brain over time.

- **Diagnostic criteria:**

A. Criteria are met for major and mild neurocognitive disorder.

B. Disturbance has insidious onset and gradual progression.

C. Disorder meets a combination of core diagnostic and suggestive diagnostic features for either possible or probable neurocognitive disorder with lewy bodies.

For probable Major or Mild neurocognitive disorder with Lewy bodies (individual has two core features or one suggestive feature)

For possible Major or Mild neurocognitive disorder with Lewy bodies (individual has only one core feature, one or more suggestive features)

1. Core diagnostic features:

a. Fluctuating cognition with pronounced variations in attention and alertness.

b. Recurrent visual hallucination that are well formed and detailed.

c. Spontaneous features of Parkinsonism

2. Suggestive diagnostic features:

a. Meets criteria for Rapid eye movement sleep behavior disorder.

b. Severe neuroleptic sensitivity.

D. The disturbance is not better explained by another neurocognitive disease or another mental disorder.

NAME : IRMA RASHEED

ROLL NO:2152

MAJOR OR MILD VASCULAR NEUROCOGNITIVE DISORDER

- **DIAGNOSTIC CRITERIA:**
- **A. The criteria are met mild or major**
- **B. Evidence of significant cognitive decline**

- 1- Cognitive deficits related to one or more carebrovascular events
- 2- Evidence for decline is prominent in complex attention
- C.Evidence of the presence of disease from history, physical examination and neuroimaging
- D. Symtoms are not better explained by another brain disease

MAJOR OR MILD NEUROCOGNITIVE DISORDER DUE TO TRAUMATIC BRAIN INJURY

- **DIAGNOSTIC CRITERIA:**
- **A. The criteria are met for mild and major**
- **B.There is evidence of a traumatic brain injury ;**
- **1- loss of consciousness**
- **2- post traumatic amnesia**
- **3- confusion**
- **4-Neurological signs**
- **C. The disorder presents immediately after the occurance of the traumatic brain injury**

MEDICATION INDUCED MAJOR OR MILD NEUROCOGNITIVE DISORDER

- **DIAGNOSTIC CRITERIA**
- **A. The criteria are met mild and major**
- **B. The impairements do not occur exclusively and persist during the medication**
- **C. The medication and duration and extent of use a capable of impairment**
- **D. The course of deficits in consistent with the timing of medication use and abstinence**
- **E. The criteria are met are not better explained by another medical condition**

NAME : SAMREEN SABA
ROLL NO :2152
Major or mild neurocognitive disorder due to hiv infection:

- **It is conitive issues caused by HIV**
- **Diagnostic criteria:**
- **A: HIV infection confirmed by medical tests.**

B: Noticeable decline in thinking and memory.

C: Daily life affected by cognitive problems.

D: Not caused by other medical conditions or substance use.

- **Major or mild neurocognitive disorder due to prion disease**
- **It's cognitive problems caused by prion disease.**
- **Diagnostic criteria:**
- **A. The criteria are met for major or mild neurocognitive disorder.**
- **B. There is indious onset, and rapid progression of impairment is common.**
- **C. There are motor features of prion disease, such as myoclonus or ataxia, or biomarker evidence.**
- **D. Do not caused by other conditions.**
- **Major or mild neurocognitive disorder due to parkinson's disease:**
- **It is cognitive issues linked to parkinson's disease.**
- **Diagnostic criteria:**
- **A. The criteria are met for major or mild neurocognitive disorder.**
- **B. The disturbance occurs in the setting of established parkinson's disease.**
- **C. There is insidious onset and gradual progression ofimpairment.**
- **D. Not caused by other coditions.**
- **Major or mild neurocognitive disorder due to Huntington's disease:**
- **It's cognitive problems associated with Huntington's disease.**
- **Diagnostic criteria:**
- **A. The criteria are met for major or mild neurocognitive disorder.**
- **B. There is insidious onset and gradual progression.**
- **C. There is clinically established Huntington's disease, based on family history or genetic testing.**
- **D. Not caused by other conditions.**
- **Mild or major neurocognitive due to another medical condition**
- **It's cognitive issues related to anther medical condition:**
- **Diagnostic criteria:**
- **A. The criteria are met for major or mild neurocognitive disorder.**
- **B. There is evidence from the history, physical examination, or laboratory findings that the neurocognitive disorder is the pathophysiological consequence of another medical condition.**
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Major or mild neurocognitive disorder due to multiple etiologies

Diagnostic Criteria:

- **Criterion A: Significant cognitive decline from previous functioning, evidenced by concern from the individual, informant, or clinician, and impaired cognitive performance.**

- Criterion B: Interference with independence and daily functioning.
- Criterion C: Not exclusively occurring during delirium; the disorder persists beyond delirium episodes.

Etiology (Causes):

- Alzheimer's Disease (AD): It's characterized by the accumulation of beta-amyloid plaques and tau protein tangles in the brain, leading to neuronal damage and cognitive decline.
- Vascular Neurocognitive Disorder: This results from impaired blood flow to the brain, often due to strokes or small vessel disease.
- Lewy Body Dementia: It's characterized by the presence of abnormal protein deposits (Lewy bodies) in the brain.
- Frontotemporal Neurocognitive Disorder: This involves the degeneration of nerve cells in the frontal and temporal lobes of the brain.
- Parkinson's Disease Dementia: Individuals with Parkinson's disease may develop dementia as the disease progresses.

Diagnostic Tools:

1. Clinical Assessment: A comprehensive medical history, physical examination, and neurological assessment are conducted to evaluate cognitive function and identify any underlying conditions.Etiology (Causes):
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    - Vascular Neurocognitive Disorder: This results from impaired blood flow to the brain, often due to strokes or small vessel disease.
    - Lewy Body Dementia: It's characterized by the presence of abnormal protein deposits (Lewy bodies) in the brain.
    - Frontotemporal Neurocognitive Disorder: This involves the degeneration of nerve cells in the frontal and temporal lobes of the brain.
    - Parkinson's Disease Dementia: Individuals with Parkinson's disease may develop dementia as the disease progresses.

Treatment:

- Medication: Depending on the underlying cause, medications may be prescribed to manage symptoms and slow down disease progression. For example, cholinesterase inhibitors (e.g., donepezil, rivastigmine) are commonly prescribed for Alzheimer's disease to improve cognitive function.

- **Behavioral Interventions: Therapy, counseling, and cognitive rehabilitation techniques can help individuals cope with cognitive decline and improve quality of life.**
- **Supportive Care: Providing a supportive environment, including assistance with daily activities and safety measures, is crucial for individuals with NCDs.**

Theories:

- **Amyloid Hypothesis (Alzheimer's): Beta-amyloid plaque accumulation disrupts neurons.**
- **Tau Hypothesis (Alzheimer's, tauopathies): Hyperphosphorylated tau leads to neurofibrillary tangle formation.**
- **Cholinergic Hypothesis (Alzheimer's): Acetylcholine deficiency in basal forebrain affects cognition.**
- **Oxidative Stress and Neuroinflammation: Imbalance and immune response contribute to neuronal damage.**
- **Synaptic Dysfunction: Disruptions in synaptic transmission impair cognitive processes.**
- **Vascular Theory: Hypertension, atherosclerosis, and stroke disrupt blood flow, causing cognitive impairment.**